

4010

**B.C.A. IInd Semester (NEP)
Examination, 2024**

DATA STRUCTURE

Paper : NBCA-201

Time : 3 Hours]

[M.M. : 70

Note :- Answer any *five* questions. All questions carry equal marks.

1. Discuss the concept of pointer and structure with proper example. Consider an integer Array A[-20 40] [30 50]. Calculate the following using Row major and Column major. (Assume integer element size 4 byte and base address of array is 2000) :

(i) Number of elements in Array

(ii) Memory occupied by Array in bytes

(iii) Location of A[-10][40] [14]

2. What do you mean by Linked List ? How linked list differ from array ? Write a program to implement singly linked list with following functionalities :

(i) Insertion at beginning

(ii) Deletion at end

[14]

3. Convert the following infix expression to :

$$((A \wedge B * C)/(E * F + G) + H/I - J)$$

(i) Postfix expression

(ii) Prefix expression

[14]

4. What do you mean by Queue Data Structure ? Discuss the procedure to implement queue using stack. Write a program to implement circular queue using Array.

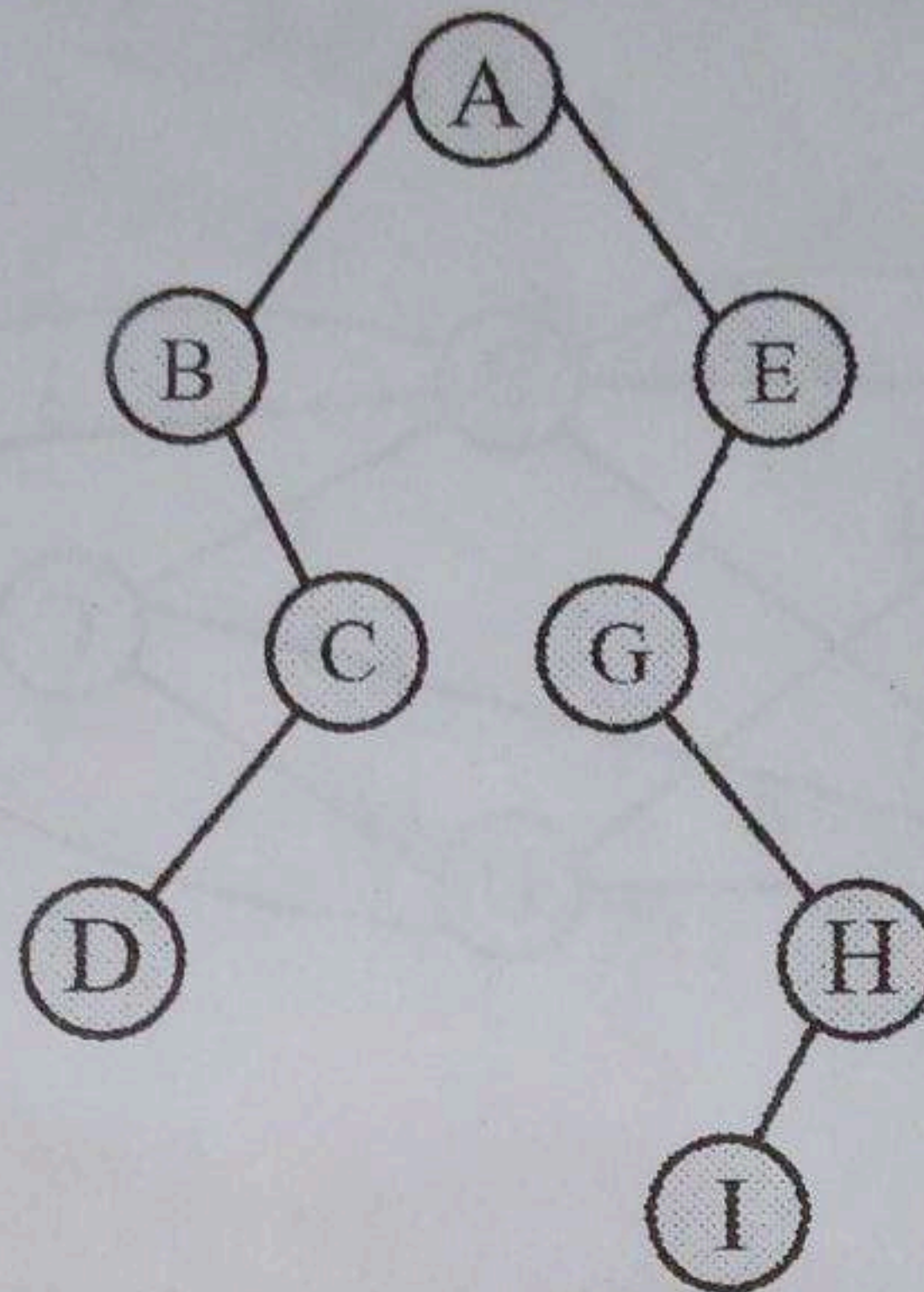
[14]

5. Define Binary search tree. How binary search tree differ from Binary tree ? Construct the binary search tree for the following pre-order :

20, 10, 5, 1, 7, 15, 30, 25, 35, 32, 40.

[14]

6. Define Binary Tree. Explain Binary Tree Representation with suitable example. Also construct the pre-order and post-order of the following Binary Tree :



[14]

7. What is Searching ? How sequential search is differ from Binary Search ? Also write the code for Binary Search.

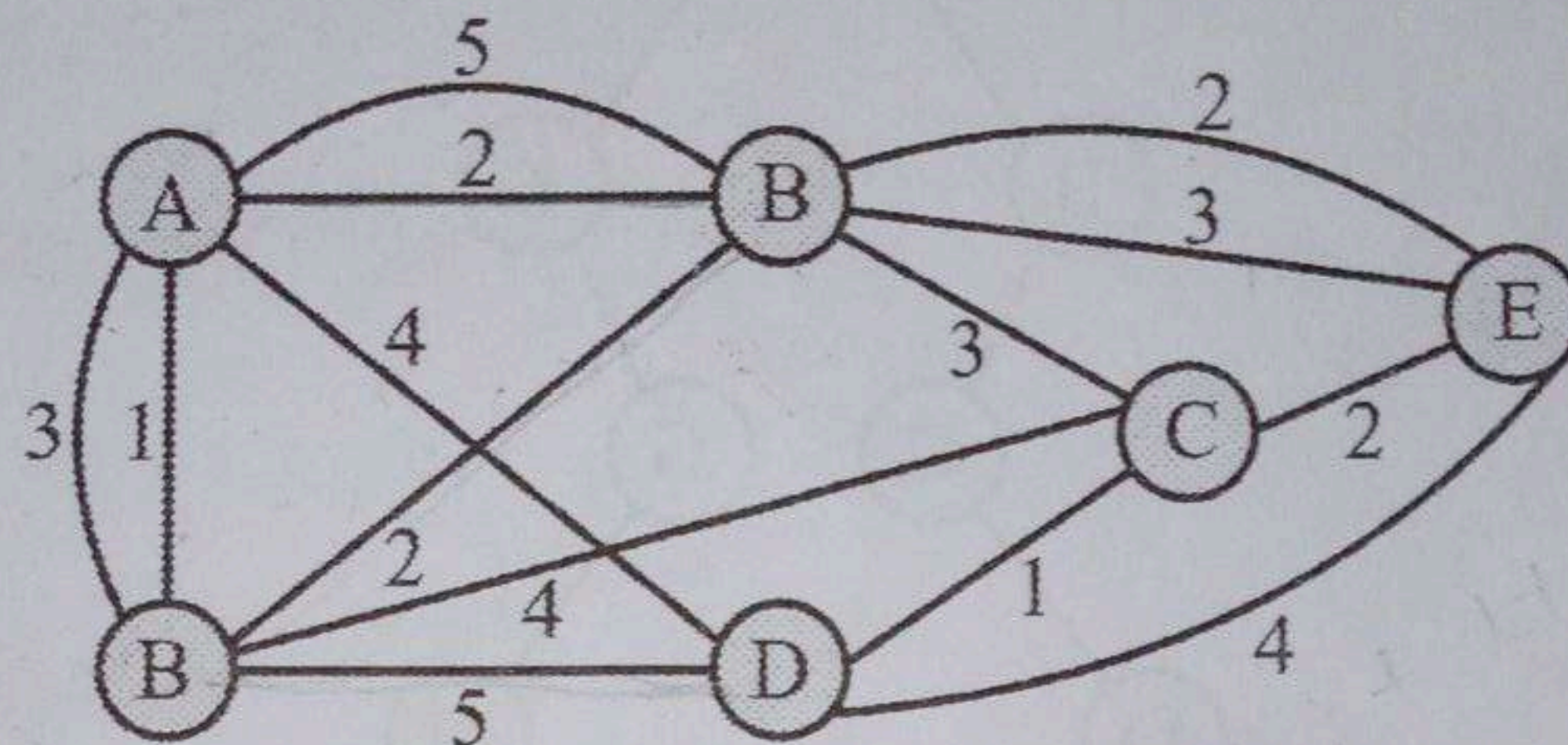
[14]

8. Explain the following with proper code :

- (i) Insertion sort
- (ii) Selection sort

[14]

9. Distinguish tree and graph with proper example and define spanning tree. Also construct the minimum cost spanning tree using Prim's algorithm of the following graph :



[14]

10. Discuss Adjacency Matrix and Adjacency list with suitable example. Also differentiate between Depth first search and Breadth first search with suitable example.

[14]